AI Coding with Huawei Cloud MaaS



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AI-Assisted Migration: Usage and Disclaimer

Sharing of capabilities, not a formal delivery commitment

Partner training | Huawei Cloud MaaS for AI-assisted coding

Central Note

This session presents the AI-assisted coding capabilities using Huawei Cloud MaaS models via LiteLLM proxy. It does not process real customer data nor constitute a formal delivery plan, security or compliance commitment, or product feature promise.

1. Data and Privacy

Security / Compliance / Protection

- No real customer data is collected or processed
- Use simulated, test, or anonymized data
- Do not upload accounts, passwords, keys, certificates, or data

2. Usage Limits

Reference / Human Review

- AI supports planning, analysis, scripts, and risk assessment
- It does not replace design, expert review, testing, or decisions
- Review plans, scripts, and configurations before the project

3. Customer Authorization

Authorization / Approval

- The customer decides whether to use AI in real projects
- Without authorization, do not enter customer information into AI
- Respect security, compliance, confidentiality, and approvals

4. Not Recommended Scenarios

Prudent Evaluation / Manual Process

- Evaluate or avoid highly sensitive security scenarios
- Classified systems, financial core, defense, or energy
- Use manual migration under strict sovereignty or AI prohibition

AI-assisted migration improves efficiency; it does not replace human review or customer authorization. Real implementation must be based on customer confirmation, security, compliance, and project situation.

The AI Coding Opportunity



Why AI Coding with MaaS?

Huawei Cloud MaaS: in-region LLM inference. Models: GLM-5.2, DeepSeek-R1, DeepSeek-Coder, Qwen2.5. OpenAI-compatible API. Data sovereignty: inference stays in-region. LiteLLM proxy: unified gateway.



Key Challenges

Model selection: which model for which task? Cost control: token budgets. Resilience: model timeouts and rate limits. Client setup: OpenCode, Claude Code, Cursor. MCP integration. Security: API key management.



AI-Assisted Approach

AI generates 200 lines Terraform in 45s vs 2 hours. Skills automate repeatable workflows. MCP tools provide real cloud data. AGENTS.md enforces conventions. Model selection: reasoning vs coding.

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Architecture & Model Routing

Architecture

Client (OpenCode/Claude Code) -> LiteLLM Proxy -> MaaS. OpenAI-compatible API. Auth: API key from ModelArts. No data leaves Huawei Cloud region. Proxy on ECS: port 4000.

Model Routing

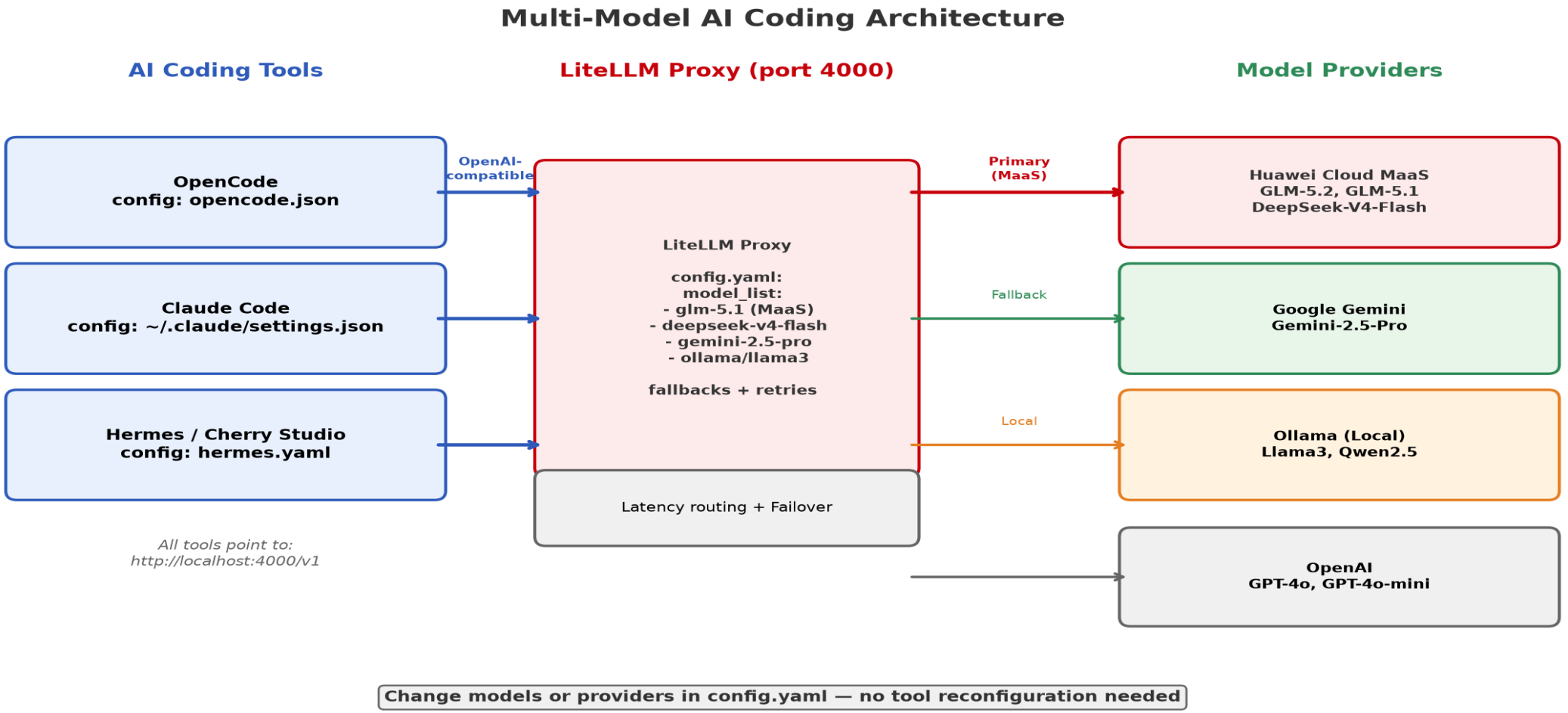
Group 1 Coding: glm-5.2 -> deepseek-coder. Group 2 Reasoning: deepseek-r1 -> glm-5.1. Group 3 Fast: deepseek-v4-flash. Group 4 Vision: qwen2.5-vl. Auto fallback on timeout.

Resilience & Fallback

Chain: primary timeout -> next in group. Retry: 3 with exponential backoff. Health check: ping MaaS every 30s. Rate limit: queue and distribute. DR: failover to another region.

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Demo: Multi-Model Architecture



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Huawei Cloud MaaS: Available Models

Ask AI: "Which MaaS model fits my task and budget?" Then verify pricing on the MaaS console.

DeepSeek-V4-Flash

1M token context window

Input: $0.135/M tokens
Output: $0.27/M tokens

Best for small_model
context compression
in OpenCode

Function call: Yes
Deep thinking: Enabled

GLM-5.1

198K token context window

Input: $0.809/M tokens
Output: $3.235/M tokens

Comparable to Claude
Opus 4.6 for coding

Function call: Yes (named)
Deep thinking: Enabled

DeepSeek-R1-0528

128K token context window

Input: $0.539/M tokens
Output: $2.156/M tokens

Complex problem solving
Reduced hallucination

Function call: Yes
Deep thinking: Always on

Source: https://www.huaweicloud.com/intl/en-us/product/maas.html | API: https://api-ap-southeast-1.modelarts-maas.com/v2/models | Region: CN-Hong Kong | Accessed: Aug 2026

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LiteLLM: Advantages for AI Coding

Example prompt: "Generate a LiteLLM config.yaml with MaaS as primary, Gemini as fallback, and Ollama for local tasks."

High Availability

If MaaS is down, LiteLLM automatically routes to Gemini or Ollama.

Fallback chains:
  primary -> next in group
Retry: 3 with backoff
Health check: every 30s

No single point of failure

Cost Optimization

small_model -> V4-Flash
  $0.135/M tokens
  60-80% cheaper

large_model -> GLM-5.1
  $3.235/M tokens
  for complex reasoning

Right model per task
= lower total cost

Single Endpoint

All tools point to localhost:4000

Change models in config.yaml:
  No tool reconfiguration
  No code changes needed
  Switch providers instantly

Context savings: 60-80%

with AI: Ask AI to generate your config.yaml with fallback chains. Verify the config syntax. Test failover by stopping MaaS and confirming routing switches.

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MaaS Models: Pricing & Characteristics

Model	Context	Input $/M tok	Output $/M tok	Deep Thinking	Func Call
GLM-5.2	198K	See console	See console	Enabled	Yes
GLM-5.1	198K	$0.809	$3.235	Enabled	Yes
GLM-5	198K	See console	See console	Enabled	Yes
DS-V4-Pro	1024K	See console	See console	Enabled	Yes
DS-V4-Flash	1024K	$0.135	$0.27	Enabled	Yes
DS-V3.2	128K	$0.27	$0.404	Disabled	Yes
DS-V3.1	128K	$0.539	$1.617	Enabled	Yes
DS-R1-0528	128K	$0.539	$2.156	Always on	Yes

All models in CN-Hong Kong region. API: https://api-ap-southeast-1.modelarts-maas.com/v2/chat/completions

RPM/TPM limits vary by account tier. See MaaS console for details. Pricing as of Aug 2026. GLM-5.1 pricing shown for input <= 32K tokens.

Source: https://www.huaweicloud.com/intl/en-us/product/maas.html | https://support.huaweicloud.com/intl/en-us/model-call-maas/model-call-019.html | Accessed: Aug 2026

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Resources & Next Steps

Resources

github.com/binrogithub/
1-3-Cloud-Adoption-Skills/
tree/main/AI-Assist-Migration/
AI-Coding

Next Steps

1. Activate MaaS (Day 1, 30 min).
2. Deploy LiteLLM (Day 1, 1h).
3. Configure client (Day 1, 15 min).
4. Add MCP (Day 2, 1h): hcloud + terraform.
5. Start coding (Day 3+).

Contact

Huawei Cloud LATAM SMB team
for MaaS setup assistance

AI Coding with MaaS

LiteLLM proxy + MCP servers + Skills
for cloud-native development.

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BENEFITS



Productivity

AI generates 200 lines Terraform in 45s vs 2 hours manual. Skills automate repeatable workflows. MCP tools provide real cloud data, not guesses. Model selection: reasoning for architecture, coding for implementation.



Cost & Sovereignty

~$30/month for typical SMB usage. Data stays in Huawei Cloud region. No external API provider needed. Right model for right task = cost optimization.



Quality & Risk

AGENTS.md enforces project conventions. Always verify: run code, tests, and lint. Memory persistence for context. Fallback chains ensure availability. Rate limit handling built-in.

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Thank you.

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